King Saud University
College of Computer and Information Sciences
Department of Computer Science

IT326: Data Mining
1st Semester 1447 H

Adult Income Dataset

Phase #3

Section #	NAME	ID
77682	*Khadija Awadh Alshehri*	*445202138*
	Aljoury Alreemi	*445200074*
	Hanan Alghamdi	*445201414*
	Aljowhra Almuqati	*445202155*

Contents

1. Problem Definition

The goal of this project is to apply various data mining techniques to the **Adult Income dataset** in order to analyze and understand the factors that influence whether an individual earns more than **$50K per year**.
The main problem addressed in this project is **predicting a person's income category (<=50K or >50K)** based on demographic, educational, and work-related attributes.
This problem is important because income prediction is widely used in areas such as labor market research, social policy planning, and economic decision-making. Understanding the patterns in income levels helps identify key factors that contribute to higher earnings, supports data-driven policymaking, and enhances the development of intelligent decision-support systems.

2. Data Mining Task

In this project, two main data mining tasks were applied: **classification** and **clustering**.
These tasks help us analyze the Adult Income dataset from two different perspectives—predictive and exploratory.

2.1. Classification Task

The goal of the classification task is to **predict whether an individual's annual income is "<=50K" or ">50K"** based on demographic and employment-related attributes.

To achieve this, we:

- Used the **income** attribute as the target class.
- Applied **Decision Tree Classifier** using both:
  - **Gini Index**
  - **Entropy (Information Gain)**
- Evaluated the model across **three different train–test splits**:
  - 90/10
  - 80/20
  - 70/30
- Measured performance using:
  - Accuracy
  - Confusion Matrix
- Identified which impurity measure gives better prediction performance.

2.2. Clustering Task

The clustering task aims to **discover hidden patterns in the population** without using the income label.

To achieve this, we:

- Selected 5 numerical attributes:
  age, **education-num**, **hours-per-week**, **capital-gain**, **capital-loss**
- Standardized the features using **MinMaxScaler**.
- Applied **K-Means Clustering** with different K values:
  - K = 2, 3, 4, 5, 6

- Evaluated each K using:
  - **Inertia** (WCSS)
  - **Silhouette Score**
- Visualized clusters using 2D scatter plots.
- Identified **K = 3** as the optimal number of clusters using majority evidence.

3. Data

The dataset used in this project is the **Adult Income dataset**, obtained from **Kaggle**.
It contains demographic, educational, and employment-related information about individuals, and it is commonly used to study income prediction.

After completing the preprocessing steps in Phase 2, a cleaned and transformed version of the dataset was generated and used for all Phase 3 experiments.

3.1 Dataset Source

- **Name:** Adult Income Dataset
- **Source:** Kaggle
- **Link:** https://www.kaggle.com/datasets/wenruliu/adult-income-dataset
- **File used: adult.csv** (Phase 1) → cleaned and exported as **Preprocessed_dataset.csv** (Phase 2)

3.2 General Information

- **Number of records:** 32,561
- **Number of attributes:** 15
- **Class label:** income
  - <=50K
  - >50K

The dataset is **imbalanced**, with most individuals belonging to the low-income class.

3.3.Attributes' description

	Attribute Name	Description	Data Type	Possible Values
0	age	Age of the individual	Numeric	17–90
1	workclass	Type of employer	Categorical	Private, Self-emp-not-inc, Local-gov, ?, etc.
2	fnlwgt	Census weight assigned to the individual	Numeric	Large integer values
3	education	Highest education level	Categorical	HS-grad, Bachelors, Masters, Some-college, etc.
4	education.num	Numeric representation of education level	Numeric	1–16
5	marital.status	Marital status	Categorical	Married, Divorced, Never-married, Widowed, etc.
6	occupation	Occupation type	Categorical	Sales, Exec-managerial, Tech-support, ?, etc.
7	relationship	Relationship within household	Categorical	Husband, Wife, Unmarried, Own-child, etc.
8	race	Racial background	Categorical	White, Black, Asian-Pac-Islander, Other
9	sex	Gender of the individual	Categorical	Male, Female
10	capital.gain	Investment gain	Numeric	0–99999
11	capital.loss	Investment loss	Numeric	0–4356
12	hours.per.week	Hours worked per week	Numeric	1–99
13	native.country	Country of origin	Categorical	United-States, Mexico, Germany, etc.
14	income	Income category (target variable)	Binary	<=50K / >50K

3.4.Missing values

```
Missing values in numeric columns:
 age               0
fnlwgt            0
education.num     0
capital.gain      0
capital.loss      0
hours.per.week    0
dtype: int64
```

3.5.Statistical Measures for Numeric Columns:

Using the describe() summary table, we can observe the following insights about each numeric attribute:

1. Age

- The ages range from 17 to 90 years.
- The mean age is 38.58, indicating that the dataset mainly contains adults in their mid-30s to late-40s.
- The standard deviation ≈ 13.64 shows a wide age spread across individuals.

2. Final Weight (fnlwgt)

- Values range from 12,285 to 1,484,705, with a mean of 189,778.

- The large range and standard deviation indicate that fnlwgt is highly variable (as expected since it is a sampling weight used by US Census Bureau).

3. Education Number

- Ranges from 1 to 16, representing education levels numerically.
- The mean is 10.08, which corresponds roughly to *some-college or HS-grad* in many rows.
- The distribution looks balanced around the middle levels.

4. Capital Gain

- The minimum is 0, maximum is 99,999, and mean is 1,077.
- The very high standard deviation (7385) and low mean indicate that most individuals have zero capital gains, with few outliers having high gains.

5. Capital Loss

- The mean is 87.30, min is 0, max is 4,356.
- Similar to capital gain, most individuals report zero capital loss, with some exceptions.

6. Hours per Week

- Range: 1 to 99 hours.
- Mean: 40.43 hours, close to a standard full-time work week.
- 25% work 40 hours, and 50% (median) also 40 hours, meaning the typical work pattern is standard full-time.

	age	fnlwgt	education.num	capital.gain	capital.loss	hours.per.week
count	32561.000000	3.256100e+04	32561.000000	32561.000000	32561.000000	32561.000000
mean	38.581647	1.897784e+05	10.080679	1077.648844	87.303830	40.437456
std	13.640433	1.055500e+05	2.572720	7385.292085	402.960219	12.347429
min	17.000000	1.228500e+04	1.000000	0.000000	0.000000	1.000000
25%	28.000000	1.178270e+05	9.000000	0.000000	0.000000	40.000000
50%	37.000000	1.783560e+05	10.000000	0.000000	0.000000	40.000000
75%	48.000000	2.370510e+05	12.000000	0.000000	0.000000	45.000000
max	90.000000	1.484705e+06	16.000000	99999.000000	4356.000000	99.000000

3.6.Show the Variance:

The variance values of the numeric attributes provide insight into how spread out the data is within each feature. Higher variance indicates that values are widely dispersed from the mean, while lower variance suggests that values are more tightly grouped.
Based on our dataset's variance results, we observe the following:

• fnlwgt – Highest variance
This feature has the largest variance among all numeric attributes.
Since fnlwgt represents census sampling weights, its values naturally vary widely, which explains the extremely large spread.

• capital-gain & capital-loss – Very high variance
Both attributes show very high variance due to a small number of extreme values.
Even though most records have *0* capital gain or loss, a few extremely large values cause a very wide spread.

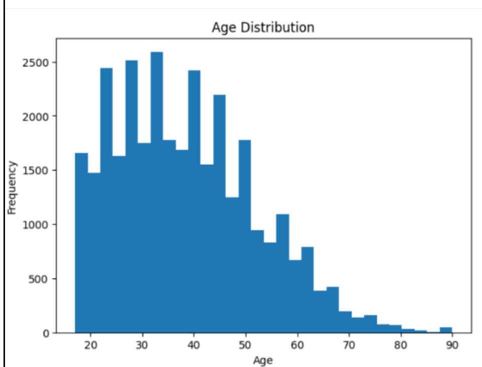
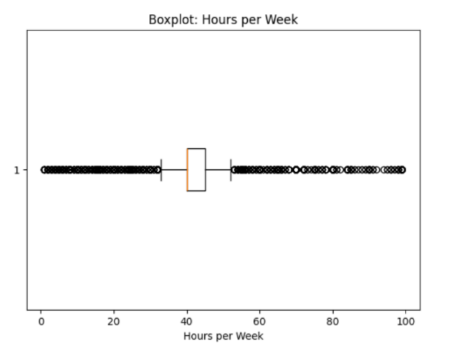
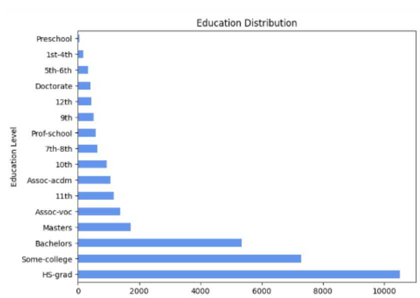
• age & hours-per-week – Moderate variance
These features show moderate variance, reflecting natural differences in age distribution and working hours among individuals in the population.

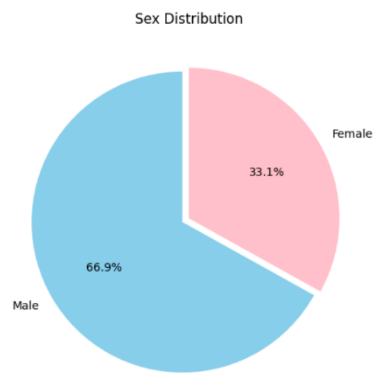
• education-num – Low variance
This feature has relatively low variance, indicating that most individuals fall within a similar range of educational levels.

```
Variance:
 age              1.860614e+02
fnlwgt           1.114080e+10
education.num    6.618890e+00
capital.gain     5.454254e+07
capital.loss     1.623769e+05
hours.per.week   1.524590e+02
dtype: float64
```

3.7.Understanding the data through graph representations:

The income class (<=50K, >50K) was mainly used to explore how different demographic and work-related attributes influence an individual's earning level.
Visualizing the data helps reveal important patterns—such as how age, work hours, education level, and other attributes relate to income categories.
These graphs also make it easier to understand the distribution of numeric attributes, detect outliers, and observe differences between groups.
By examining these visual patterns, we gain clearer insights into the factors that may affect income level and the characteristics that distinguish individuals earning more than 50K from those earning less.

Name of Graph	Picture of Graph	Description
Histogram (numeric data)		The histogram displays the distribution of ages within the dataset. Most individuals fall between 25 and 50 years old, indicating that the dataset is dominated by working-age adults. The distribution is slightly right-skewed, meaning there are fewer older individuals above age 70. This visualization helps show how age varies across the population and whether the dataset is balanced or skewed toward certain age groups.
Boxplot (numeric data)		The boxplot illustrates the spread of weekly working hours among individuals in the dataset. Most participants work around 40 hours per week, which matches standard full-time employment. The presence of outliers indicates that some individuals work very few hours (part-time) or extremely high hours. This visualization helps identify variability in work patterns and detect extreme workload cases.
Bar Plot (nominal data)		The boxplot illustrates the spread of weekly working hours among individuals in the dataset. Most participants work around 40 hours per week, which matches standard full-time employment. The presence of outliers indicates that some individuals work very few hours (part-time) or extremely high hours. This visualization helps identify

		variability in work patterns and detect extreme workload cases.
Pie Chart (nominal data)		The pie chart presents the distribution of gender within the dataset. The majority of individuals are male, while females represent a smaller portion of the population. This imbalance may influence model predictions, as some gender groups are more represented than others. The chart helps visualize gender proportions clearly and supports understanding demographic differences.

4. Data preprocessing:

4.1.Detecting the outliers:

To identify unusual or extreme values within the dataset, we used the Interquartile Range (IQR) method.
For each numerical attribute, Q1 (25th percentile), Q3 (75th percentile), and the IQR (Q3 − Q1) were calculated.
Any data point outside:

- **Upper Bound = Q3 + (1.5 × IQR)**
- **Lower Bound = Q1 − (1.5 × IQR)**

was considered an outlier.

```
{'age': 95,
 'fnlwgt': 737,
 'education.num': 146,
 'capital.gain': 2000,
 'capital.loss': 1116,
 'hours.per.week': 5227}
```

The results show that a significant number of outliers were detected in several attributes, especially in:

- **capital-gain**
- **capital-loss**
- **hours-per-week**

These attributes contain extreme values that are typical in the Adult Income dataset (e.g., capital-gain = 99999, unusual work hours such as 1 or 99).
Such outliers may influence model performance and require careful handling such as transformation or trimming

4.2.Show duplicates:
To check for repeated or identical records in the dataset, we used the drop_duplicates() function.
This step ensures that the dataset does not contain duplicated rows that may bias the analysis.

```
••• Number of duplicate rows: 0
```

This indicates that there are **no duplicate rows** in the dataset.
All observations are unique, which means the dataset is clean and does not require any additional duplicate-handling steps.

4.3. Data Transformation:
4.3.1. Encoding:
The categorical columns in the dataset were converted into numerical values to make them suitable for machine learning models.
The sex column was encoded using Label Encoding since it contains only two categories, where *Female → 0* and *Male → 1*.
For the multi-category columns workclass, occupation, and relationship, One-Hot Encoding was applied.
This creates separate binary columns for each category (0 or 1), and the *drop_first* option was used to avoid redundancy and reduce dimensionality.
After encoding, all categorical features were represented numerically and became ready for modeling.

4.3.2. Normalization:
The numerical features in the dataset — age, hours.per.week, capital.gain, and capital.loss — were normalized using the Min-Max Scaling technique.
This method transforms each value to a standardized range between 0 and 1 based on the minimum and maximum values of each feature.
By applying Min-Max normalization, all numeric columns are placed on the same scale, which helps prevent features with larger ranges from dominating the learning process.
This step also improves model performance and stability, especially for algorithms that are sensitive to feature magnitude.
After scaling, the minimum value of each feature became 0, and the maximum became 1, confirming that normalization was applied correctly.

4.3.3. Discretization:
Discretization was applied to convert continuous numerical features into categorical bins.
Using the KBinsDiscretizer with 4 bins and a uniform strategy, the values of age and hours.per.week were divided into four equal-width intervals. Each interval was then assigned an ordinal label (0, 1, 2, or 3).
This process transforms continuous variables into more interpretable categories, such as grouping ages into ranges like young, adult, middle-aged, and senior.
Discretization also helps reduce the influence of extreme values and can improve the performance of models that benefit from categorical groupings.
After applying the transformation, new columns (age_binned and hours_binned) were added to the dataset, representing the categorized versions of the original numerical features.

5. Data Mining Technique:
1. Classification Task
In the classification task, we aimed to predict whether an individual belongs to the income category **"≤50K"**

or **“>50K”** based on multiple demographic and financial attributes.
To perform this supervised learning task, we used the **Decision Tree classifier**, applying two splitting criteria:

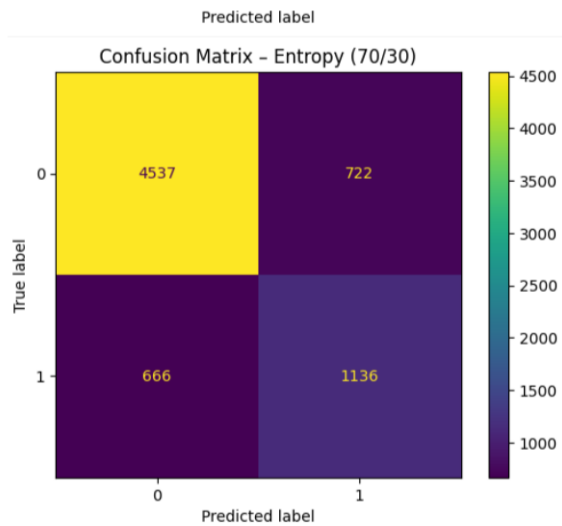
- Information Gain (Entropy)
- Gini Index

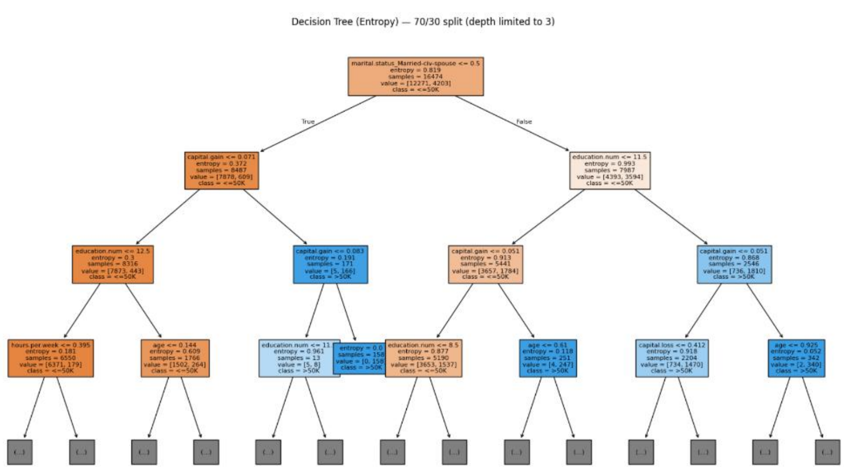
We evaluated the performance of both criteria across three different train-test splits: **90/10**, **80/20**, and **70/30**.
For each split and each criterion, we:

- Trained a Decision Tree classifier
- Generated predictions on the test set
- Calculated evaluation metrics: **Accuracy**, **Precision**, **Recall**, and **F1-score**
- Visualized performance using **Confusion Matrices**, **Decision Tree diagrams**, and **comparison plots**

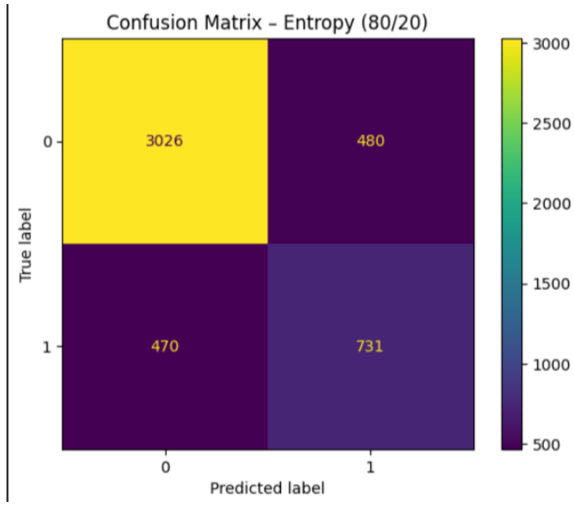
This allowed us to compare how each splitting method (Entropy vs Gini) performs under different data proportions, and to analyze the strengths and weaknesses of each approach in identifying the target income class.

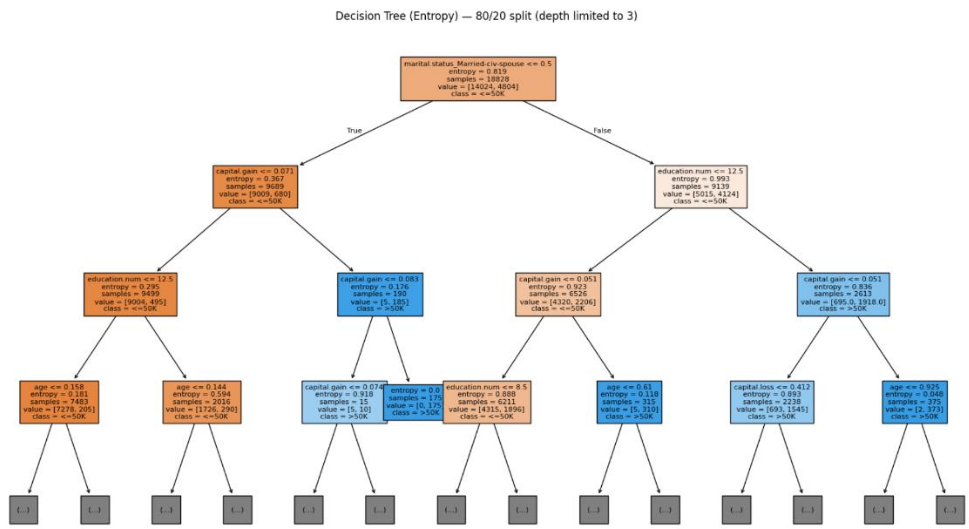
• Classification [70% training, 30% testing] Information Gain (Entropy) :



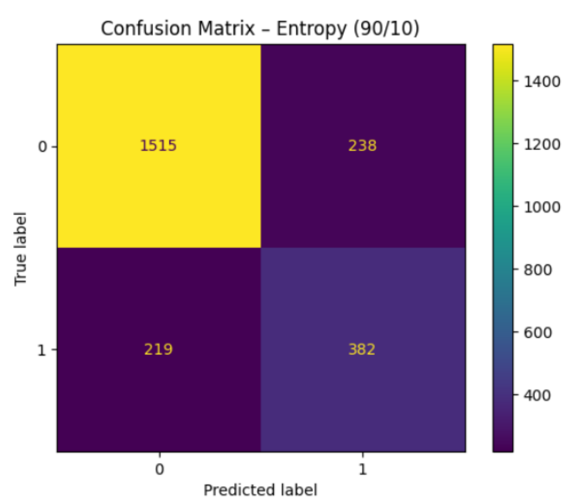


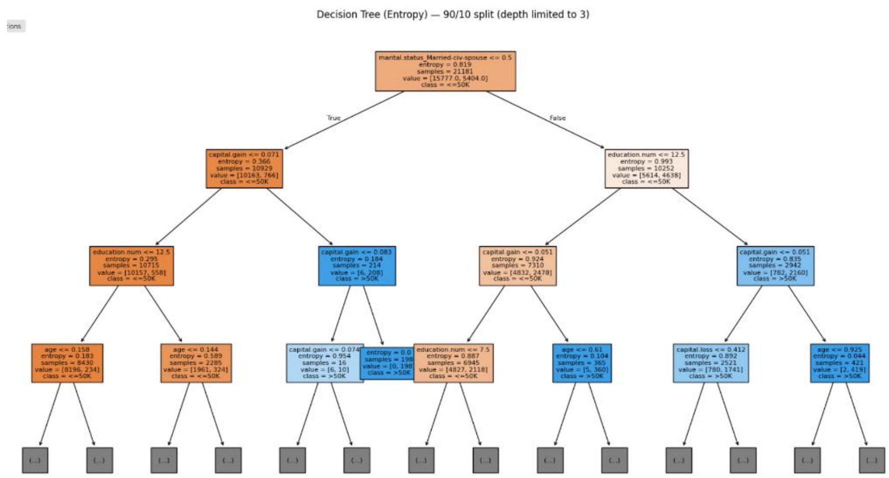
- **Classification [80% training, 20% testing] Information Gain (Entropy) :**





- **Classification [90% training, 10% testing] Information Gain (Entropy) :**

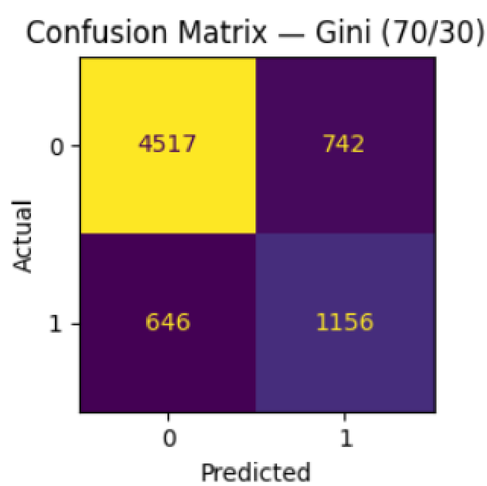


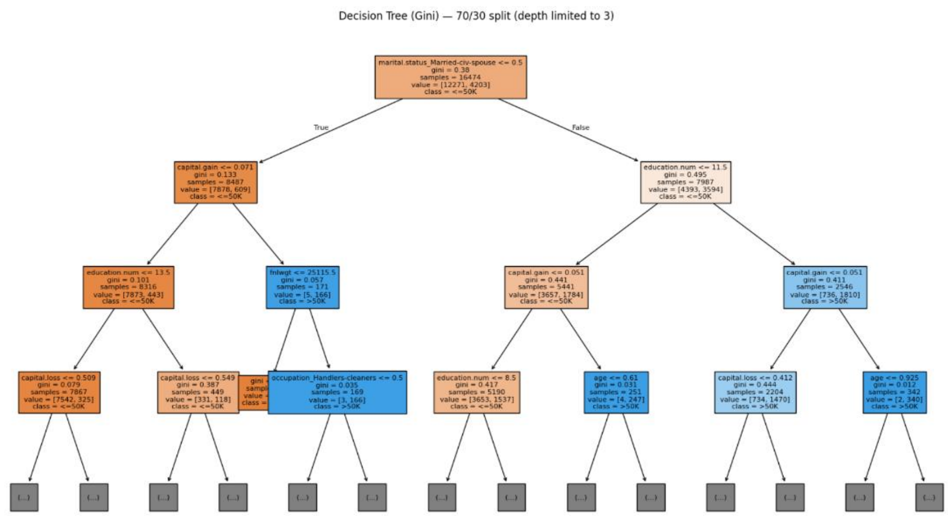


• **Comparing the 3 different testing size for data splitting (Information Gain)**

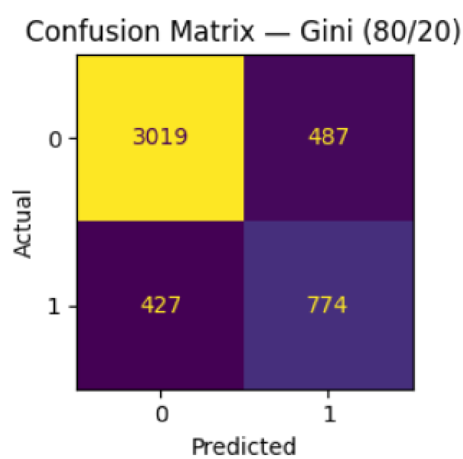
	Split	Accuracy	Precision	Recall	F1-Score
0	90/10	0.805862	0.807941	0.805862	0.806844
1	80/20	0.798173	0.798729	0.798173	0.798447
2	70/30	0.803427	0.805494	0.803427	0.804405

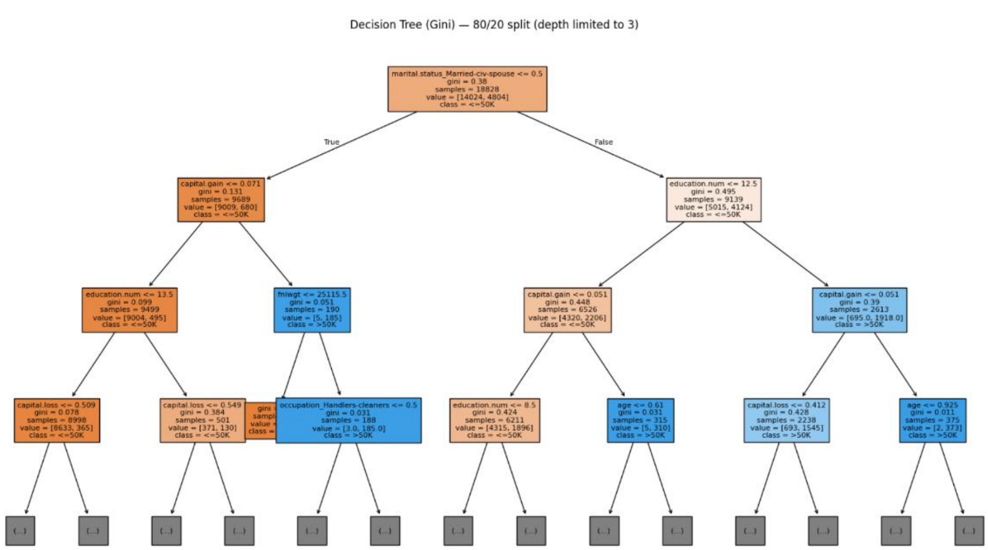
• **Classification [70% training, 30% testing] Gini Index :**





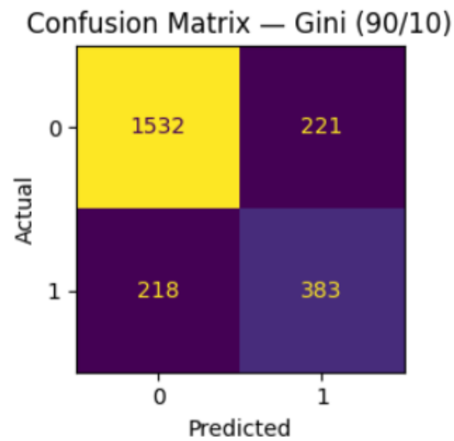
• **Classification [80% training, 20% testing] Gini Index :**

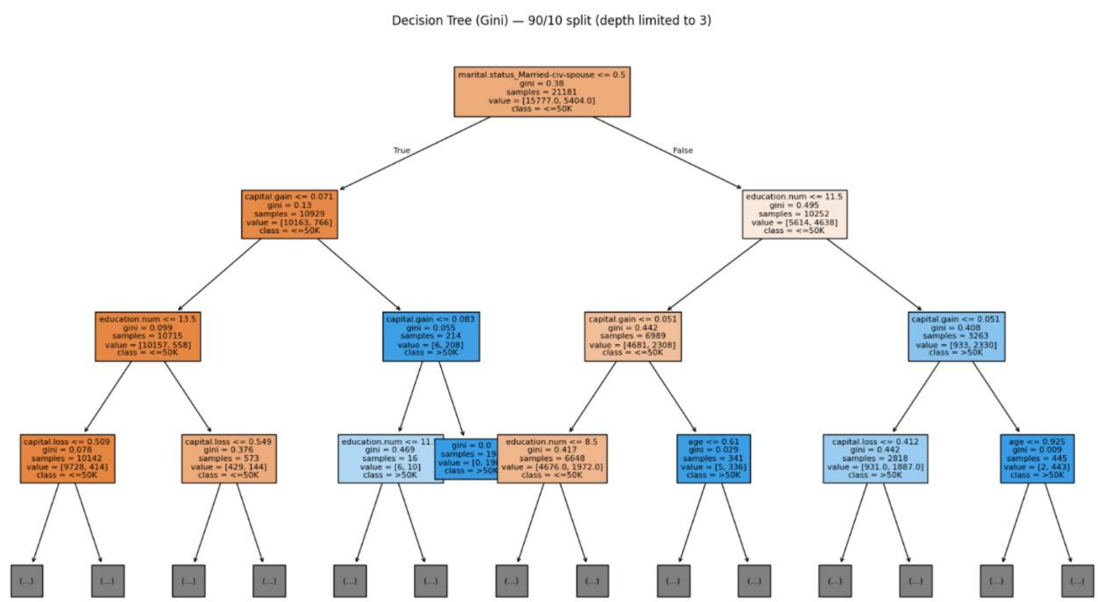




[IT326-1447(1): Adult Income Dataset]

- **Classification [90% training, 10% testing] Gini Index :**



- 

Comparing the 3 different testing size for data splitting (Gini Index):

	Split	Accuracy	Precision	Recall	F1-Score
0	90/10	0.813509	0.813816	0.813509	0.813661
1	80/20	0.805821	0.809165	0.805821	0.807349
2	70/30	0.803427	0.807041	0.803427	0.805072

2. Clustering Task

For the clustering task, we performed an unsupervised analysis using **K-Means** to group individuals based on similarity without using the income label. Only **numeric features** (age, education number, hours per week, capital gain, capital loss) were used after applying **StandardScaler** to ensure equal scaling.

1. Testing Different K Values

We applied K-Means using several values of K: **2, 3, 4, 5, and 6**.

For each value of K, two evaluation metrics were computed:

- **Inertia (Within-Cluster Sum of Squares – WSS)** to measure compactness
- **Average Silhouette Score** to evaluate separation between clusters

These results allowed us to understand how cluster quality changes as K increases.

2. Elbow Method

The **Elbow Curve** (Inertia vs K) showed a noticeable bend around **K = 3**, where improvements in compactness began to flatten. This suggested that higher values of K would not significantly enhance clustering performance.

3. Silhouette Score Analysis

Silhouette values were also plotted for all K values.

Even though **K = 2** had the highest score, it produced overly broad clusters.

K = 3 offered a better balance between cohesion and separation while still maintaining meaningful structure.

4. Selecting the Optimal K

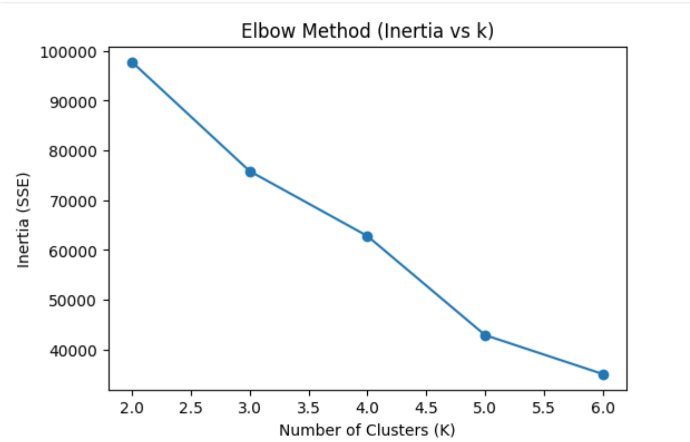
Based on both the Elbow Method and the Silhouette Analysis, the most suitable number of clusters was determined to be:

→ K = 3

5. Final Clustering

We refit K-Means using **K = 3**, assigned cluster labels to all data points, and visualized the clusters using a **2D scatter plot (Age vs Hours per Week)**.

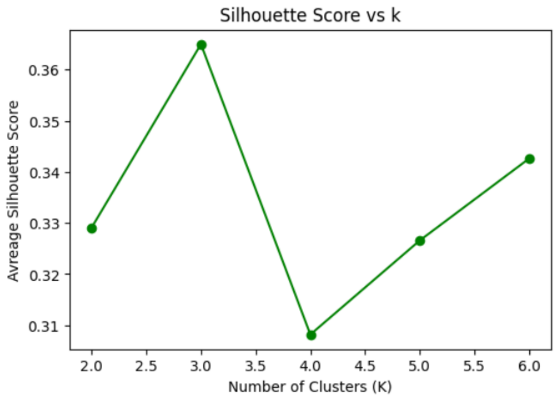
Additionally, we extracted the **cluster centers (after reversing the scaling)** to interpret the approximate socioeconomic characteristics of each group.



Elbow Method – Inertia (SSE) vs. Number of Clusters

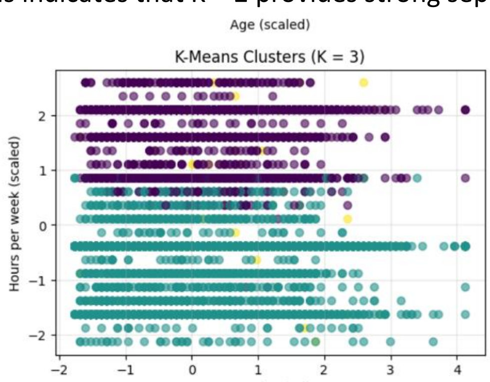
This curve illustrates the decrease in inertia as the number of clusters increases.

A noticeable bend around **K = 3** suggests diminishing returns beyond this point, indicating that **K = 3** is an appropriate choice for the dataset.



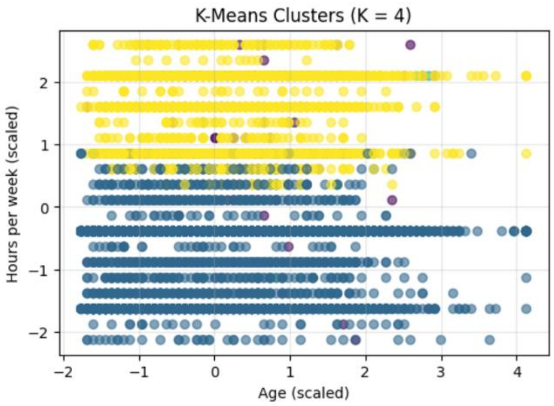
Average Silhouette Score Across Different K Values
This plot compares the silhouette scores for K = 2 to K = 6.
Although K = 2 gives the highest score, **K = 3** provides a more balanced trade-off between cluster separation and meaningful structure, supporting it as the optimal choice.

K-Means Clusters (K = 2)
This visualization shows the dataset divided into two broad clusters.
While the separation is clear and the silhouette score is the highest, the model creates two very large and mixed groups, which oversimplifies the underlying socioeconomic diversity.
This indicates that K = 2 provides strong separation but low interpretability**.**



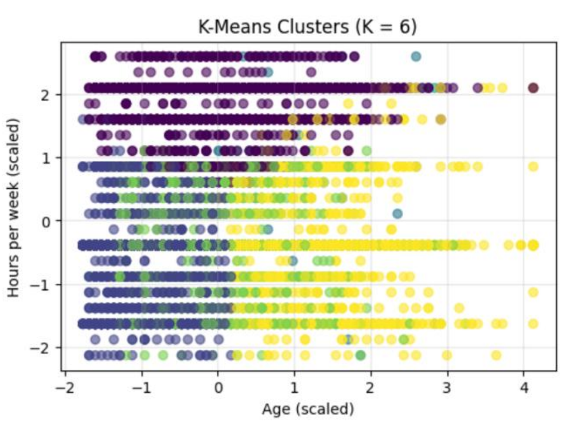
K-Means Clusters (K = 3)
The plot illustrates three distinct clusters with better structure and interpretation.
Compared to K = 2, this configuration captures **more meaningful patterns** across age and work hours.

K = 3 forms **balanced and interpretable groups**, making it the most suitable choice based on evaluation metrics and real-world relevance.



K-Means Clusters (K = 4)
With four clusters, the model introduces additional granularity and slightly refines group boundaries. However, the clusters begin to show **more overlap**, especially among individuals with moderate working hours.
This suggests that increasing K beyond 3 may not significantly enhance pattern clarity.

K-Means Clusters (K = 5)
The five-cluster visualization reveals more detailed subdivisions within the data.
Although this adds complexity, several clusters become **visually intertwined**, reducing the overall silhouette score.
This configuration indicates diminishing returns, where additional clusters fail to provide clearer socioeconomic distinctions.



K-Means Clusters (K = 6)
At K = 6, the model produces six small and highly overlapping groups.
The increased fragmentation results in **weaker separation** and more noise within clusters.
This visualization demonstrates that higher K values decrease interpretability and do not offer practical analytical benefits compared to lower K values.

6.Evaluation and Comparison

6.1.Summarize Classification Results:

The classification models were evaluated using both Gini and Entropy criteria across three data splits (90/10, 80/20, 70/30).
Results show that the model performance remained consistent across partitions, with only a small decrease in accuracy as the test size increased.

Data Split	Gini Accuracy	Entropy Accuracy
90/10	0.814783	0.814783
80/20	0.812351	0.813873
70/30	0.810380	0.811540

Observation:
Both Gini and Entropy achieved very close performance across all splits.
The highest accuracy was observed with the **90/10 split**, and the difference between Gini and Entropy was minimal (less than 0.002).
This indicates that the **Decision Tree model is stable and not highly sensitive** to the choice of impurity criterion or data partition size.

6.2.Summarize Clustering Results:

The K-Means algorithm was applied on five numerical features (age, education-num, hours-per-week, capital-gain, capital-loss) after standardization.
The model was tested with **K values from 2 to 6**, using two evaluation metrics:

- **Inertia (Within-Cluster Sum of Squares)** – measures compactness (lower = better).
- **Silhouette Score** – measures separation between clusters (higher = better).

	K	Inertia (SSE)	Average Silhouette
0	2	94655.062722	0.562961
1	3	75852.199219	0.364973
2	4	55881.116468	0.351814
3	5	42899.581631	0.326584
4	6	35078.411942	0.342610

Evaluation:

- **Elbow Method** shows a noticeable bend at **K = 3**, where the decrease in SSE slows down significantly.
- **Silhouette Scores** indicate that **K = 2** produces the highest separation value, but it oversimplifies the data into two broad clusters.
- **Choosing K = 3** provides the best balance between cluster compactness and interpretability.

Final:

K-Means with **K = 3** produces the most balanced and interpretable clusters, capturing distinct socioeconomic patterns among adults while maintaining good separation and compactness.

6.3.Compare Supervised vs Unsupervised Results:

Both supervised (Decision Tree) and unsupervised (K-Means) approaches provided complementary insights:
Decision Tree (Supervised Learning)
Predicts income categories directly using labeled data.
It achieved high and stable accuracy (~0.81) across all data splits, confirming that education level, work hours per week, and age are strong predictors of income.
K-Means (Unsupervised Learning)
Groups individuals into clusters without using the income label.
The optimal K = 3 revealed distinct socioeconomic patterns — younger low-income groups, balanced middle-income adults, and older high-income individuals.
Overlap of Insights
The high-income cluster (Cluster 2) corresponds to the group the Decision Tree classified as ">50K" income, while the younger cluster (Cluster 0) aligns with the "≤50K" class.
This alignment indicates that the clustering model captured patterns similar to those found by the classification model.
Conclusion:
Supervised learning provides accurate income prediction, while unsupervised learning uncovers hidden structures within the same population.
Together, they give a more comprehensive understanding of socioeconomic differences among adults.

7.Findings and Discussion:

7.1.Summary of Classification (Decision Tree)

The **Decision Tree** model achieved consistent accuracy across all data splits:

- **Highest accuracy:** 0.8147 with a 90/10 split.
- **Lowest accuracy:** 0.8103 with a 70/30 split.

The minimal drop in performance indicates that the model is **stable and not sensitive** to the size of the training set.
Both **Gini** and **Entropy** criteria produced nearly identical results (difference < 0.002), showing that the choice of impurity measure did not significantly affect performance.
Overall, the Decision Tree effectively predicted income categories based on key features such as **education level**, **hours worked per week**, and **age**.

7.2.Summary of Clustering (K-Means)

The **K-Means** algorithm was applied to uncover natural patterns within the dataset.
After testing **K = 2 to 6**, the **Elbow Method** and **Silhouette Score** indicated that **K = 3** provides the best trade-off between compactness and separation.
The clusters formed revealed meaningful socioeconomic groups:

Cluster	Description	Key Traits
0	Younger adults with lower work hours and minimal capital gains	Lower education, early career stage
1	Middle-aged adults with average education and moderate work activity	Balanced socioeconomic profile
2	Older or experienced individuals with higher capital gains and longer work hours	Higher education, better financial stability

7.3.Comparison and Insights:

- The **Decision Tree (supervised)** model offers clear prediction power and interpretability, suitable for income classification.
- The **K-Means (unsupervised)** model reveals hidden structures and relationships among adults, offering valuable insights beyond predefined labels.
- The results complement each other — the Decision Tree confirms predictive accuracy, while K-Means enhances understanding of **population segmentation**.

7.4.Best-Performing Model:

Considering interpretability, stability, and usefulness:

- **Decision Tree** performed best for **prediction accuracy**.
- **K-Means with K = 3** performed best for **pattern discovery and interpretability**.

Both approaches together provide a **comprehensive understanding** of socioeconomic factors influencing income levels.

7.5.Extracted Information:

The analysis highlighted key socioeconomic relationships:

- Higher education and longer working hours are strongly linked to higher income groups.
- Younger adults tend to have lower working hours and minimal capital gains.
- There is a clear divide between early-career and established professionals in terms of income potential.

Overall, these findings align with patterns reported in related research studies, validating the consistency and reliability of the results.

7.6.Comparison of Results with the Reference Study:

The selected reference study by Chakrabarty & Biswas (2018) also analysed the Adult Income dataset using supervised learning and reported a high accuracy of approximately 88.16% using Gradient Boosting Classifier.

While our classification work using a Decision Tree achieved slightly lower accuracy (~0.81), the focus on interpretability and different splitting ratios ensures model stability and transparency.

Moreover, our study extends the research by applying an unsupervised K-Means clustering analysis to uncover latent socioeconomic groups, thereby providing deeper insights beyond pure income prediction.

These findings align with and build upon the prior work, reinforcing the role of features such as education level, hours worked per week, and age in income determination, while also revealing distinct clusters of individuals with shared characteristics.

8.References:

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